



WeatherGPT

Next-Generation AI Meteorological & Disaster Intelligence Platform

A comprehensive, multi-lingual, AI-powered weather forecasting and natural disaster decision-support system featuring 2-way real-time voice intelligence, AI cloud vision (Weather Lens), localized agriculture advisories, and interactive 3D WebGL radar telemetry.

Project: WeatherGPT Full-Stack
AI

Domain: AI / WebGL / Atmospheric
Intelligence

Status: Production-
Ready

1. Executive Summary

WeatherGPT is a state-of-the-art meteorological intelligence platform engineered to bridge the critical divide between complex numerical atmospheric science and actionable human decision-making. While conventional weather applications present disconnected metrics (humidity, dew point, atmospheric pressure) requiring manual user interpretation, WeatherGPT utilizes Large Language Model (LLM) reasoning and Computer Vision to deliver proactive, domain-specific guidance.

The platform provides hyper-localized agriculture advisories (Kisan Mitra), road safety transit assessments, coastal and cyclone hazard intelligence for fishermen, real-time AI sky camera classification (Weather Lens), and an interactive 2-way voice assistant supporting English and 22 scheduled Indian languages.

2. System Architecture & Engineering Design

WeatherGPT is architected as a decoupled, high-performance client-server infrastructure built for ultra-low latency and resilience against cloud serverless timeouts:

- **Client Architecture (SPA):** Built with React 19, TypeScript, and Vite. Utilizes Tailwind CSS for glassmorphism theming and Three.js for 60 FPS 3D hardware-accelerated rendering.
- **Direct Client-to-Backend Telemetry:** All AI chat, voice synthesis, and vision inference requests bypass intermediate serverless proxies and connect directly from the user's browser to the dedicated Render backend API, entirely preventing 10-second serverless timeout terminations during long LLM inferences.
- **Isolated User Scoping:** Chat histories, user settings, and preferences are strictly partitioned per authenticated Google OAuth email (`weathergpt_sessions_v3_<email>`) to preserve absolute data privacy.

3. Technology Stack & Frameworks

Layer	Technologies	Role in Architecture
Frontend Core	React 19, TypeScript, Vite	Single Page Application (SPA), component lifecycle, type-safety
Styling & UI	Tailwind CSS, Lucide Icons, Glassmorphism	Responsive deep navy dark theme, frosted glass cards
3D Graphics	Three.js (WebGL), THREE.Timer	Interactive 3D Weather Globe, AI Voice Orb, Particle canvas
Mapping & GIS	Leaflet, React-Leaflet, OpenStreetMap, Nominatim	Interactive precipitation radar, click-to-estimate rainfall
Speech & Audio	HTML5 Audio, Web Audio, Web Speech API	Real-time voice dictation, neural TTS, progressive word highlight
Backend Service	Node.js / Express on Render	Meteorological data integration, Gemini AI LLM inference
Deployment & CI/CD	Vercel (Client), Render (API), GitHub	Continuous integration, automated builds, zero-downtime deploys

4. Detailed Module Specifications

4.1 Multi-Lingual AI Voice Assistant & Live Call

Provides a conversational 2-way voice interface. As the AI speaks, the `SpokenTextRenderer` dynamically highlights the active word in real time and smoothly auto-scrolls down to track speech progression. When the call is exited or modal closed, `stopAllAudio()` instantly halts speech synthesis and audio playback to prevent audio leaks.

4.2 Weather Lens (AI Sky & Cloud Camera)

Leverages Computer Vision to analyze cloud photographs uploaded or captured by user mobile cameras. The model classifies cloud formations (Cumulonimbus, Altostratus, Cirrus), measures cloud density percentage, and calculates instant rain probabilities.

4.3 Interactive Precipitation Radar & Map Telemetry

Displays interactive layers for precipitation, temperature, wind gusts, and cloud cover. Clicking any coordinate calculates localized rainfall estimates and reverse-geocodes the exact location name via Nominatim.

4.4 Kisan (Agriculture) Advisory Engine

Tailored agricultural intelligence for farmers. Analyzes soil moisture, evaporation rates, and rainfall windows to recommend optimal dates for pesticide spraying, fertilizer application, and crop harvesting.

4.5 Travel Weather Planner & Road Hazard Index

Calculates transit safety scores between departure and arrival cities, warning drivers in advance about crosswinds, low visibility, and flood-prone highway zones.

5. Indian Regional Language Support (22+ Languages)

To ensure maximum accessibility across rural and urban populations, WeatherGPT natively supports full UI and voice localization across 22 official Indian languages:



6. API Endpoints & Telemetry Reference

Endpoint Route	Method	Payload / Params	Description
/api/ask	POST	{ question, location, conversationId }	Primary LLM weather intelligence inference endpoint
/api/weather	GET	?city=Rajkot	Fetches current, hourly, and 7-day meteorological forecast
/api/weather/lens	POST	{ imageBase64, question, location }	Vision analysis for cloud type & precipitation risk
/api/weather/compare	POST	{ locations: ["Rajkot", "Surat"] }	Simultaneous multi-city weather comparative analytics
/api/voice/speak	POST	{ text, language }	Neural text-to-speech audio stream generation
/api/disaster/alerts	GET	?location=Gujarat	Fetches active severe weather emergency advisories

7. External Evaluation & Viva Q&A Guide

Q1. What makes WeatherGPT superior to conventional weather apps?

Conventional apps display raw meteorological data points without real-world context. WeatherGPT integrates LLM contextual reasoning to translate numeric measurements into actionable guidance (e.g., whether to spray crops, travel safety indices, and multi-lingual voice guidance).

Q2. How did you resolve cloud serverless timeouts for AI voice responses?

Serverless platforms (like Vercel) enforce a strict 10-second execution cap. We eliminated intermediate proxy hops by routing client fetch calls directly to our dedicated Render backend service, utilizing HTML5 Audio and Blob ObjectURLs for real-time binary streaming.

Q3. How is user privacy and chat history managed across multiple logins?

Sessions are keyed dynamically to the authenticated Google OAuth email identifier (`weathergpt_sessions_v3_<email>`). When a user logs in or switches accounts, the client strictly retrieves and updates only their isolated session records.

Q4. How is the real-time spoken word karaoke highlight accomplished?

The `SpokenTextRenderer` breaks text into token streams. During speech synthesis or HTML5 audio playback, word boundary events track character offsets, highlight active tokens with glowing pill styling, and trigger smooth auto-scrolling to keep the reading position centered.

WeatherGPT Project Documentation • Live URL: weather-gpt-tau.vercel.app • Repository: github.com/sharaddetroja/WeatherGPT-